I. Set Theory (GSE 518)

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In order to understand the concepts and methods that we will discuss in this class, it is important to have an underlying understanding of some key concepts from Set Theory.

1 Set Definitions

A set is a collection of objects.

For two sets A and B:

- A is a subset of B (denoted $A \subset B$) if and only if (IFF) all elements of A are also in B. ($\forall x \in A \implies x \in B$)
- A = B IFF $A \subset B$ *and* $B \subset A$
- Let $\emptyset$ denote the empty set
  - Note that $\emptyset$ is a subset or element of every set

2 Basic Operations With Sets

We restrict attention to the Universal Set Ω.

$\rightarrow$ All set operations are done relative to Ω

$\rightarrow$ All sets we use are subsets of Ω

Example:

let $\Omega = \{1, 2, 3, 4\}$

$A = \{1, 2\}$

$B = \{1, 3, 4\}$

A and B are subsets of Ω

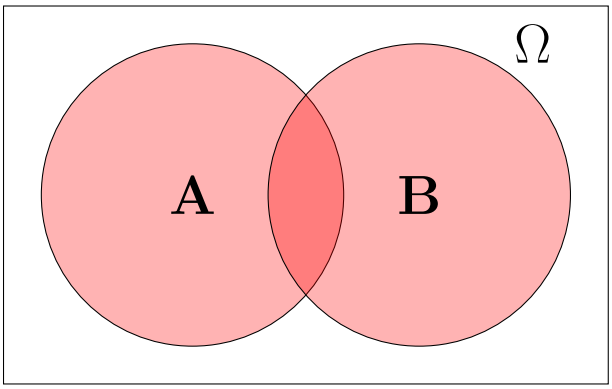
Definitions:

For two sets A and B in Ω ($A, B \subset \Omega$),

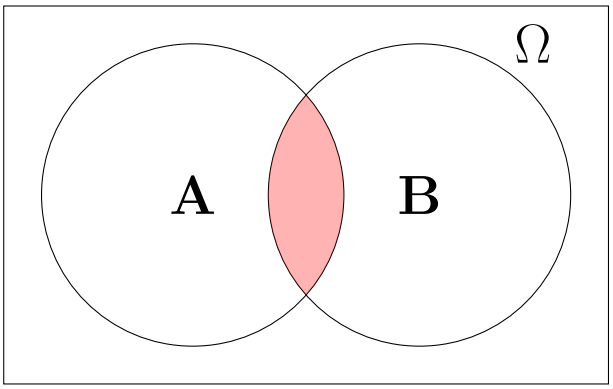
- Union : $A \cup B$ = set of all x in A *or* B
- Intersection : $A \cap B$ = set of all x in A *and* B
- Complement : A^c , $\overline{A}$ = set of everything not in A still in our universal set Ω

  Note: $\Omega^c = \emptyset$, $\emptyset^c = \Omega$, $A \cup A^c = \Omega$

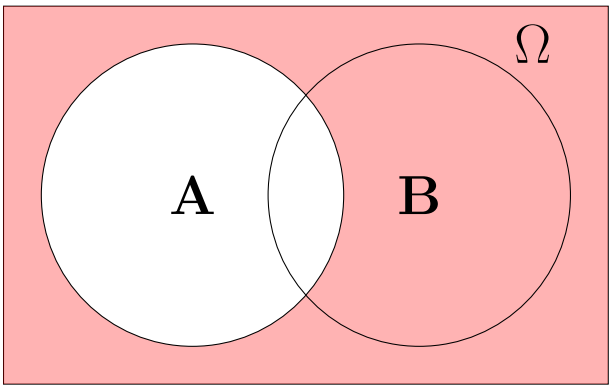
$A \cup B$



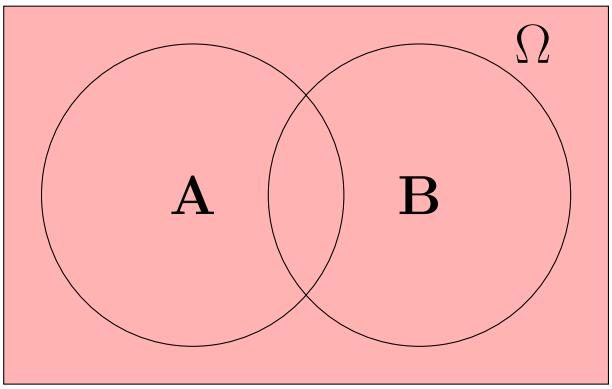
$A \cap B$



A^c



$A \cup A^c$



3 Basic Set Laws

Theorem:

For any 3 sets A, B, C defined in Ω:

a) Commutative law:

$$A \cup B = B \cup A$$
$$A \cap B = B \cap A$$

b) Associative law:

$$A \cup (B \cup C) = (A \cup B) \cup C$$
$$A \cap (B \cap C) = (A \cap B) \cap C$$

c) Distributive law:

$$A \cap (B \cup C) = (A \cap B) \cup (A \cap C)$$
$$A \cup (B \cap C) = (A \cup B) \cap (A \cup C)$$

d) DeMorgan's law:

$$(A \cup B)^c = A^c \cap B^c$$
$$(A \cap B)^c = A^c \cup B^c$$

Next: Extension to *collection* of sets.

4 Uncountable and Countable Sets

Definitions:

Let A be a set, and define $|A|$ = number of elements in A.

We define A as a finite set if $|A|$ is finite.

We define A as an infinite set if $|A|$ is not finite.

We define A as countable if $|A|$ is finite *or* $|A| = |\mathbb{N}|$.

— This means we can define a 1 to 1 function mapping from $A \to \mathbb{N}$.

We define A as uncountable if A is not countable.

Examples:

1) $c = \{x : x \in \mathbb{N} \text{ and } x < 100\} = \{1, ..., 99\}$

— Since c is a finite set, c is countable.

2) $c = \{x : x \in \mathbb{N} \text{ and x is even}\} = \{2, 4, 6, ...\}$

— c is an infinite set, but countably infinite set since a one to one mapping can be made from c to $\mathbb{N}$.

3) $c = \mathbb{R}$

— c is infinite and uncountable since $\mathbb{R}$ has more elements and therefore is "more infinite" than $\mathbb{N}$.

→ *Intuition:* Consider the discrete subset of $\mathbb{N}$, $\{1, 2\}$. This is a finite set with two elements. Consider the continuous subset of $\mathbb{R}$, [1,2]. This continuous set contains an infinite number of elements. So between every natural number 1,2,3,... in $\mathbb{R}$, there exists infinitely many elements of $\mathbb{R}$.

5 Collection of Sets

Definitions:

Let $\{A_1, A_2, A_3, ..., A_n\}$ be a finite collection of sets in Ω. Then:

$\bigcup_{i=1}^{n} A_i = A_1 \cup A_2 \cup ... \cup A_n$
$\bigcap_{i=1}^{n} A_i = A_1 \cap A_2 \cap ... \cap A_n$

DeMorgan's Application:

$(\bigcup_{i=1}^{n} A_i)^c = \bigcap_{i=1}^{n} A_i^c$
$(\bigcap_{i=1}^{n} A_i)^c = \bigcup_{i=1}^{n} A_i^c$

Let $\{A_1, A_2, A_3, ...\}$ be an infinite but countable collection of sets in Ω. Then:

$\bigcup_{i=1}^{\infty} A_i = \{x \in \Omega : x \in A_i, \text{for some } i \in \mathbb{N}\} = A_1 \cup A_2 \cup A_3...$

→ This means if x is in $\bigcup_{i=1}^{\infty} A_i$, x is present in at least one A_i.

$\bigcap_{i=1}^{\infty} A_i = \{x \in \Omega : x \in A_i, \forall i \in \mathbb{N}\} = A_1 \cap A_2 \cap A_3...$

→ This means if x is in $\bigcap_{i=1}^{\infty} A_i$, x is present in all A_i's.

Note: These ideas can be extended to an infinite and uncountable collection of sets as well.

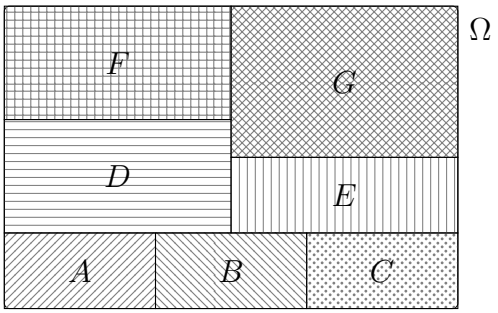
6 Partitions

Definitions:

Two sets A,B $\in \Omega$ are (pairwise) disjoint or mutually exclusive if $A \cap B = \emptyset$.

If the sets $A_1, A_2, ...$ are pairwise disjoint and $\bigcup_{i=1}^{\infty} A_i = \Omega$ then the collection of sets form a partition of Ω

Example of a partition:



Example:

Let Ω be all individuals in a population.

Let $A_1 = \{\text{males}\}$, $A_2 = \{\text{females}\}$.

If $A_1 \cap A_2 = \emptyset$ and $A_1 \cup A_2 = \Omega$ then

A_1, A_2 form a partition of Ω.

Example:

Let $\Omega = [1, \infty)$ and $A_i = [i, i+1)$ for $i \in \mathbb{N}$.

Does $A_1, A_2, \ldots$ form a partition of Ω?

1. Check to make sure disjoint:

   $A_i \cap A_j = \emptyset$ for all $i, j \in \mathbb{N}$ with $i \neq j$.

2. Check to make sure they form Ω

   $\bigcup_{i=1}^{\infty} A_i = [1,2) \cup [2,3) \cup [3,4) \ldots = [1, \infty)$

Thus, $A_1, A_2, \ldots$ form a partition of Ω.

7 Concluding Results

Definition:

The complement of B relative to A is:

$$A\backslash B = \{x \in \Omega : x \in A \text{ and } x \in B^c\} = A \cap B^c$$

Final Results: (Proofs)

1. $A = A\backslash B \cup (A \cap B)$

   Proof. Let $A, B \in \Omega$.

   $$\begin{aligned} & A\backslash B \cup (A \cap B) \\ \Leftrightarrow\ & A \cap (B^c \cup (A \cap B)) \end{aligned}$$

   then, by Distributive Law:

   $$\begin{aligned} \Leftrightarrow\ & A \cap (B^c \cup B) \\ \Leftrightarrow\ & A \cap \Omega = A \end{aligned}$$

   □

2. If $B \subseteq A$ then $A = A\backslash B \cup B$

   Proof.

   $$\begin{aligned} & A\backslash B \cup B \\ \Leftrightarrow\ & (A \cap B^c) \cup B \\ \Leftrightarrow\ & (A \cup B) \cap (B^c \cup B) \\ \Leftrightarrow\ & A \cap \Omega = A \end{aligned}$$

   (since $B \subseteq A, A \cup B = A$) □

3. $A \cup B = A \cup (B \cap A^c)$

   Proof.

   $$\begin{aligned} & A \cup (B \cap A^c) \\ \Leftrightarrow\ & (A \cup B) \cap A \cup A^c \\ \Leftrightarrow\ & (A \cup B) \cap \Omega \Leftrightarrow A \cup B \end{aligned}$$

   □